

Kiew Yao Hong

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SUMMARY & SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, NestJS, Node.js, Express, FastAPI

Frontend & Tools: React, Astro, HTML, CSS, TailwindCSS, Docker, Pivotal Cloud Foundry (PCF), AWS, Git

EXPERIENCE

DBS Bank

Jul 2024 – Present

Software Engineer

- Backend engineer in Cards and Payments domain, developing Spring Boot microservice middleware supporting secure integration between core banking systems and external vendors (VisionPLUS, PowerCARD).
- Delivered two migrations of mainframe and vendor-dependent systems to a cloud-native platform, increasing in-house ownership for debit and credit card systems, reducing operational overhead and long-term costs.
- Implemented RESTful APIs to serve bank wide business stakeholders and retail consumers ensuring compliance while integrating payment workflows.
- Performed root-cause analysis using PCF application logs to resolve production and integration issues, reducing incident resolution time by 30% while adhering to strict incident management and change control processes
- Led end-to-end change request management for 10+ production changes monthly in a regulated environment, coordinating technical inputs and cross-regional approvals

NCS Group

Nov 2022 – Jun 2024

Software Engineer

- Modernized legacy systems to cloud-native architecture on AWS, improving reliability, maintainability, and regulatory compliance under Whole of Government (WoG) initiative.
- Delivered full-stack web application using React and Spring Boot, leveraging AWS CloudWatch monitoring to reduce incident response time by 50% while strengthening SLA compliance for government systems.
- Refactored RESTful API endpoints and optimized database queries, significantly improving API performance and system responsiveness under production load for serving government clients.
- Led the redesign of React components into reusable, modular patterns, consolidating over 20 static pages into a configurable dynamic module and significantly reducing code duplication while improving developer productivity and maintainability.
- Proposed and implemented system enhancements aligned with evolving client needs, improving solution adaptability and contributing to long-term revenue growth.

PROJECTS AND MISCELLANEOUS

Expense Application | React, Java, Docker, PostgreSQL

- Started as a teaching project for an online developer community and evolved into a fully functional finance platform through iterative development.
- Implemented transaction workflows with consistent frontend–backend state synchronization, experimenting with asynchronous event processing to explore system decoupling.

Pathfinding Visualiser | Python, Pygame

- Built interactive grid-based graph visualizer supporting dynamic wall placement and start/end node configuration.
- Implemented Dijkstra visualization to animate node exploration and shortest distance

YouTube Comment Analyser | Python, FastAPI, React, TypeScript

- Developed a web tool assisting viewers to quickly understand the sentiment and discussion trends of any YouTube video without manually reading through hundreds of comments.
- Implemented summarisation via Groq's model to generate concise insights about audience sentiment and discussion themes.

EDUCATION

University of Wollongong

Computer Science, Cybersecurity

Singapore Institute of Management

Sep 2020 – Mar 2022

Temasek Polytechnic

Diploma in Information Technology

Apr 2015 – May 2018